

NeuroLearn

A privacy-first study tool for neurodivergent students

Dyslexia · ADHD · ASD

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Scan for source code

Abstract

NeuroLearn turns pasted notes into structured summaries for higher-education students with dyslexia, ADHD or ASD.

It pairs a structured summariser with calm reading modes, text-to-speech with line highlighting, and a focus timer with micro-tasks, all inside one privacy-first workflow.

Existing tools like Otter, Notion, Grammarly and Glean cover parts of the journey but stay fragmented and rarely combine accessibility, attention support and GDPR controls in one place. NeuroLearn closes that gap.

Introduction & Background

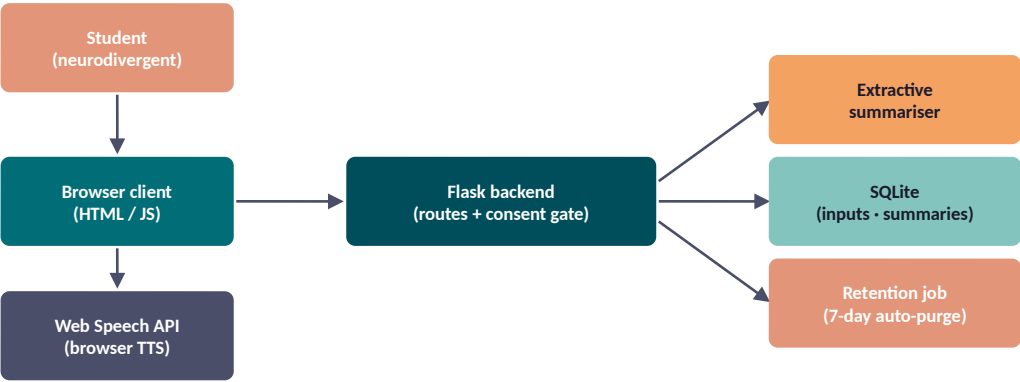
Around 10 to 15 percent of UK students identify as neurodivergent and mainstream study tools place a heavy cognitive load on them through busy interfaces, weak typography and split workflows.

Students typically need to chain multiple apps together, transcription in one, summarising in another, focus timers in a third, which fragments attention and makes it harder to retain what was just read.

NeuroLearn addresses this by integrating structured summaries, low-stimulus and high-contrast modes, multimodal output via TTS and attention scaffolding through a Pomodoro-style timer, all grounded in WCAG 2.2 AA and operationalised UK GDPR (consent, minimisation, 7-day retention).

Diagram / Design

System architecture and key user flows



Privacy is enforced at the API boundary: consent before processing, data minimisation, 7-day auto-deletion and a one-tap delete-my-data control.

Four reading modes (R2)

Default Standard contrast Aa The quick brown fox	Low-stimulus Muted, no animation Aa The quick brown fox
High-contrast WCAG AA, focus rings Aa The quick brown fox	Dark high-contrast Yellow-on-black Aa The quick brown fox

Core flow: Paste notes → Structured summary → Read aloud with line highlighting → Focus session → Delete my data

Specification & Implementation

Stack: Python 3 + Flask, SQLite, vanilla HTML/CSS/JS, Web Speech API for TTS. No external LLM API used in the MVP, the summariser runs entirely on-device.

Requirements implemented (MoSCoW Musts and Should):

- R1 Paste notes → structured summary (sentence-scoring extractive summariser).
- R2 Four reading modes plus font size and line spacing sliders, persisted in DB.
- R3 TTS via Web Speech API with line-level highlighting and 0.5x to 2x speed.
- R4 Pomodoro focus timer (25 min default) with editable micro-task checklist.
- NFR-A WCAG 2.2 AA contrast, keyboard focus order, reduced-motion support.
- NFR-P Consent banner, one-tap Delete my data, hourly 7-day auto-purge job.

Testing & Evaluation

Targets and methods

≥ 80% Task completion (3 core tasks)	≥ 70 SUS score (n ≥ 5 pilot)	≤ 30 s Latency for 2-5 min input
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Methods

- Small pilot (n ≥ 5) running three tasks: paste → summary, toggle reading mode/TTS, start a focus session. Time-on-task, errors and SUS logged.
- Accessibility checked with axe / Lighthouse, plus manual keyboard passes and contrast checks. Each failure maps to a fix ticket.
- Latency measured against the ≤ 30s target on 2-5 minute pasted inputs.

Limitations

- Small pilot, single demo user (no full auth), audio-input path deferred (text-only fallback per Part 1 rationale), and the extractive summariser is intentionally lighter than an LLM to keep the system runnable on a low-spec VM.

Conclusions & Future Work

Contribution

NeuroLearn shows that accessibility, attention support and privacy-by-design can sit alongside a working summariser inside a single workflow rather than scattered across multiple apps.

The MVP delivers every Must requirement (R1, R2, R3, NFR-A, NFR-P) plus the focus timer Should (R4). The technical decisions, a text-first pipeline, SQLite for simple retention, an on-device summariser and Web Speech API for TTS, keep the system runnable on a lower-spec VM and remove the cost and quota risks of relying on external APIs.

By integrating the four pillars (calm reading modes, structured summaries, multimodal output, attention scaffolding) inside one privacy-first flow, NeuroLearn closes a gap that fragmented tools like Otter, Notion and Glean leave open for neurodivergent learners.

Future work

- Audio input with on-device speech-to-text so the full transcribe → summarise → read flow works end-to-end.
- Personalised disability profiles so accessibility settings adapt automatically to dyslexia, ADHD or ASD presets rather than being toggled manually each time.
- A larger pilot study recruited through university accessibility services, plus a comparison against an LLM-based summariser to evaluate quality versus latency tradeoffs.
- Cross-device sync of preferences and saved summaries while keeping retention defaults intact.
- Optional integration with university VLEs (Moodle, Canvas) so lecture material can be pulled in directly without manual pasting.